



What are locks, OpLocks and delayed breaks in ONTAP?

https://kb.netapp.com/on-prem/ontap/da/NAS/NAS-KBs/What_are_locks_OpLocks_and_delayed_brea...

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Applies to

- ONTAP 9
- CIFS / SMB
- NFS

Answer

Normal Locks	Stops other client machines from accessing file while an existing client is accessing it. Cannot be removed.
Oplocks	Allows for sharing of read access of file and client side caching of locked information.

- Oplocks, here called locks, are set by a client on write access and must be released to allow another client exclusive write access, and caches some, or all of the file to the client machine for editing.
- Oplocks allow a client to cache data locally to increase performance and reduce network use. However, if a CIFS session is lost, any changes that have not been written to disk can be lost.
- Oplocks can be broken manually or based on rules defined in ONTAP

Example sequence:

1. Client1 opens `\\storage_system\share\file1` requesting a batch or exclusive oplock
2. Storage system responds with a batch or exclusive oplock for file1 to Client1
3. Client2 attempts to open `\\storage_system\share\file1` requesting a batch or exclusive oplock
4. Storage system holds off on the open request to Client2 and sends an Oplock Break Request to Client1, requesting it to flush/commit its changes and release its locks.
5. Client1 responds to the Oplock Break Request, flushing/committing its cached writes
6. Storage system grants the open to Client2 with the appropriate lock

Note: In Step 4, if Client1 does not respond to the Oplock Break Request in 35 seconds, the storage system does three things:

1. Prior to Data ONTAP 9.0 Log, an Oplock Delayed Break message that includes the IP Address of the offending client to the syslog
2. Forcefully cleans up any locks associated on the file for Client1
3. Grants the open response to Client2

Additional Information

- [How to break locks via ONTAP CLI](#)
- ["The document is locked for editing by another user" error not presented when a file is opened after another user has the file open](#)